

AutiStudy: An AI-Powered Virtual Shadow Teacher for Autistic Students in Pakistan

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Abstract—Autism Spectrum Disorder (ASD) students have a learning disorder, and in Pakistan, most schools still do not have proper support for these learners, especially in STEM subjects. There is a concept of a shadow teacher in Pakistan, but they are very expensive and not available for all students. This paper presents a solution AutiStudy, a web-based AI learning platform that is designed to work like a virtual shadow teacher for ASD students. It has bilingual support and uses multi-modal features (text, images and voice), so students can understand lessons simply and comfortably. In traditional AI systems, outdated information is present, not the curriculum-aligned information, so AutiStudy uses Pakistan-Curriculum Textbooks, where the enhanced RAG pipeline is used for General Science and Computer Science. The RAT (Retrieval Augmented Thinking) pipeline is used for Mathematics. The results clearly show better performance in our system than in older systems. The platform also includes practice quizzes, fun learning activities and a parent-teacher dashboard to track student progress. Overall, AutiStudy shows that AI can support ASD learners in a simple, practical and affordable way, especially in countries like Pakistan where learning resources are limited and costly.

Index Terms—Autism Spectrum Disorder, Adaptive Learning, Retrieval Augmented Generation, Retrieval Augmented Thinking, RAG, RAT, Large Language Models, Special Education, Bilingual AI, Pakistan

I. INTRODUCTION

The World Health Organisation estimates that about 1 in 100 children worldwide are diagnosed with Autism Spectrum Disorder (ASD). In South Asia, including Pakistan, the awareness of this disorder is slowly growing among people. Children with Autism are not less intelligent, they just have a different learning style with different learning needs due to this, they need a learning environment that matches how they understand and feel. They have different issues, such as focusing for long, having difficulty in understanding the abstract concepts, and handling too much sensory input, so they need repeated and consistent teaching.

In Pakistan, the main way to support autistic students in school is to provide them with a shadow teacher. The shadow teacher sits with the children and helps them during the class. While this idea is helpful, there are major issues, such as there are not enough trained teachers, the cost of the shadow teachers is too high for many families, and many of them just focus on life skills and communication. So there is still a gap where autistic students are not learning STEM subjects, and there is no bilingual language facility.

The paper presents a solution named AutiStudy, a web-based AI learning platform for autistic students that will provide simple step-by-step explanations of complex STEM subjects: science, math, and computer science. The system is designed to be patient, adaptive, and easy to understand using text, voice and visuals. It also supports bilingual language, Urdu and English, helping students learn independently without needing a full-time shadow teacher.

The remaining sections of the paper are as follows: Section II discusses the literature review, Section III discusses the related work, Section IV explains the methodology, Section V describes the results, Section VI shows the discussion, Section VII presents the conclusion, Section VIII discusses the future work, and Section IX gives acknowledgement to all the persons involved in this process.

II. RELATED WORK

The literature review shows a big gap between AI tools used for education nowadays and the needs of ASD students. Hajej et al. [3] use a two-phase machine learning system in their research that can detect with up to 99.99 per cent accuracy and uses an autism clarity map to recommend teaching strategies: PECS, PRT and technology-based learning using the RF-XGBoost technique. D.G. Musieva et al.'s [5] research shows that AI-based learning management systems are accepted by high school teachers, but in their system, there is a limitation

of a bilingual STEM delivery feature that our system covers. F.D et al.'s [16] research shows that autistic students show almost perfect effect sizes ($\text{Tau-U}=1.00$) for improvement in social skills and reduction in behavioural disruptions when an adaptive STEM project-based learning framework was applied. K.A.R Nelaka et al. [12] use autoencoders as the pre-processing layer in a deep learning-based facial emotional recognition system for autistic children. This system gives it 85.6 per cent accuracy, which shows that AI systems that are specifically designed for ASD students have the ability to perform better. The importance of providing different ways for autistic students to engage was highlighted by universal design learning frameworks [18], so that the students can learn while expressing themselves in an inclusive education. This framework supports our AutiStudy multi-model interface design. AuThink System by M. Dias et al. [8] was a game-based and web-based computational thinking tool that was specifically designed for autistic students according to their sensory needs, which also improves the skills of two out of three students. It does not provide STEM-based educational content like the system AutiStudy provides. In the AI section, L. Danurata et al.'s research [1] supported the RAG-based chatbots that are using textbooks as their knowledge bases. These chatbots will be able to provide more precise and more relevant answers than traditional LLMs because traditional LLMs are educated on the previous data. A. Salemi and H. Zamani [4] brought their e-RAG idea and demonstrated its usefulness. They show that the significance of retrieved documents can significantly impact the performance of an RAG system. Too many document retrievals may reduce the quality of the evaluation, and this was evident in the choice of a final context limit of $k=6$ documents for Autistudy. M. Qiang et al.'s [13] research shows that a 320 ms increase in the processing time is required to remove irrelevant documents before LLM generation, which can achieve up to 88 per cent agreement with the human evaluators. In D.M Cordova Esparza's research [7], she reviewed 82 studies and concluded that if we combine human guidance with AI learning, this could produce the best learning results instead of using just AI. This idea is also used in the Autistudy parent-teacher dashboard module. The use of similarity-based early exit and cross-encoders can speed up the system by up to 3.5 times and the use of performance does not exceed 1.2 per cent, as shown by J. Szepesvari et al. [24]. This had a significant influence on the effective re-ranking design implemented in the AutiStudy.

A. Mobile Apps For Autism Communication

Many products are currently available on the App Store and the Play Store that help autistic learners, only in the communication and initial stages of their development. For example, Mental Imagery Therapy for Autism (MITA), is a visual exercise program that uses puzzle-like components to improve language and focus on skills in children under 7. In autistic learners, gamification as a learning platform is used in early social skills, communication, and initial math skills. In Turkey, Otsimo is a company that develops learning apps

for autistic learners. Applications that are considered Augmentative and Alternative Communication (AAC), like GoTalk NOW and Proloquo2Go, help non-verbal or minimally verbal children build sentences using pictogram boards. Although these applications are designed well for their specific goals, they have some weaknesses: they fail to support upper primary STEM learning, they are costly, and they do not offer content in Urdu. Due to this, in Pakistan, most people do not easily have access. AutiStudy solved all of the weaknesses.

B. General-Purpose AI Tutoring Platforms

General AI tutoring tools such as Khan Academy's Khanmigo, Socratic by Google and Photomath are effective for explaining learning material for different grades and subjects. But these platforms are made for neurotypical learners. They used standard reading levels, do not provide sensory adjustments settings and little support for different learning speeds for neurodivergent learners. Large language model AI assistants like Gemini and ChatGPT can answer questions, but sometimes they give complex and general answers. Responses are not always age-appropriate or fully aligned with the school curriculum.

C. AI in Pakistani Educational Technology

In recent years, Pakistan's EdTech sector has progressed quickly. Platforms like Taleemabad, Sabaq Foundation and Maqsd are helping students with low-cost online learning, mostly through video lessons based on the national curriculum. These platforms work well for regular students, but they are not designed for learners who need extra support and want to learn in different ways. Some governments provide tablet-based learning programs in Punjab and Khyber Pakhtunkhwa, but they are still focused on general classroom students and not designed for autistic learners. According to our research, Pakistan still doesn't have an AI system specifically designed for autistic students at the upper primary STEM level. This shows a clear gap, which AutiStudy aims to fulfil. Globally, AI learning tools are increasing after ChatGPT entered the market. Platforms such as Khanmigo and Magic-School use AI to support learning, but they are mainly built for general English-speaking students. They also rely on broad AI knowledge and are not closely aligned with school textbooks, which makes them less suitable for autistic learners who need clear, simple step-by-step guidance. As shown in Table I, AutiStudy brings together autism friendly design, curriculum-based learning, bilingual support, accurate textbook-based answers and step-by-step reasoning in one simple system.

III. METHODOLOGY

A. System Architecture Overview

Our web-based platform, AutiStudy, is divided into two parts: frontend and backend. The frontend is developed on Next.js and React 18 UI Library, whereas the backend is developed with the help of FastAPI and Flask. For the embedding generation, we used Sentence Transformers(allmini-LM-L6v2), and for vector embedding storage, we use ChromaDB,

TABLE I
COMPARISON OF RELATED SYSTEMS

Feature	MITA	Otsimo	Khan Academy	ChatGPT T	AutiStudy
STEM (Grade 4-7)	No	Partial	Yes	Yes	Yes
Autism-adapted	Yes	Yes	No	No	Yes
Urdu support	No	No	No	Partial	Yes
Curriculum-aligned	No	No	Yes	No	Yes
Hallucination-free	N/A	N/A	Partial	No	Yes
Multi-agent AI	No	No	No	No	Yes
Emotion detection	No	No	No	No	Yes
Step-by-step math	No	No	Yes	Partial	Yes

where all the embeddings of the textbook are stored. The system also has an AI layer in which a hybrid RAG Pipeline is used, in which there is a combination of Dense + BM25 + Cross-Encoder reranker + RRF (Reciprocal Rank Fusion). OpenAI GPT 4o Mini is used for text generation. For image generation, we use Dall-E and the GPT Image model; for voice output, we use the OpenAI TTS model. The web application can be used on any standard web browser.

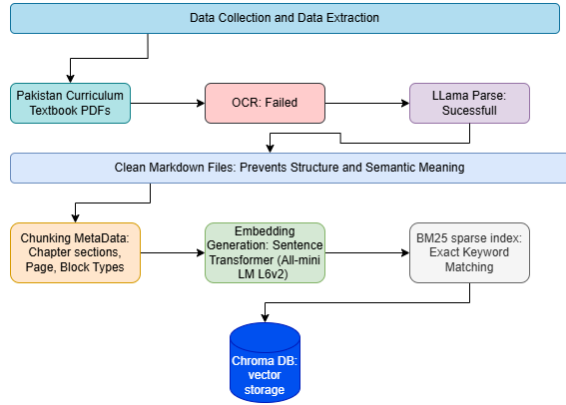


Fig. 1. Data Collection and Extraction

B. Data Collection and Content Preparation

The first step in the methodology was the data collection. The Pakistan curriculum textbooks have been used as the data in our study, which is available in the PDF format. We have selected STEM subjects: math, science, and computer science. In order to extract the content of the textbook, we use a technique called optical character recognition (OCR), but it gives us low-quality results. For example, the mathematical equations, symbols of fractions, Urdu text, and tables were not in the proper format. We changed our technique and then used LLama Parse, which is an AI-driven document parsing technique by LLama Index. LLama Parse maintains the document structure, mathematical notation, headings, and formatting when converting the textbook's contents to clean Markdown files. After extracting the content, it is divided into meaningful chunks instead of fixed-sized token segmentation. Each chunk has the following metadata: Chapter Name,

Section Title, Page Number and Block Type. The block type is made up of Definition, Procedure, Example, Practice and Explanation. Also, to prioritise the retrieval of the procedural and explanatory content, a metadata weighting technique was applied.

C. Embedding Generation and Vector Storage

Once the chunks are prepared, the semantic embeddings are obtained by passing them through the sentence transformer model all-mini-lm-l6v2. For the storage of the embeddings, we have used ChromaDB, which served as a better database than other databases, and it gives us efficient similarity-based retrieval. Additionally, the BM25 sparse retrieval method was used to get accurate keyword matching for exercise numbers, textbook terms, and subject-specific phrases, which dense retrieval alone would not be able to capture.

D. RAG With Enhanced Hybrid Retrieval Pipeline and Neural Re-Ranking

The system then uses a hybrid retrieval pipeline for improving the retrieval procedure and accuracy, and discarding irrelevant responses. The pipeline starts when a user asks a query from the AutiStudy chatbot. The system first uses intent detection to detect the category of the question. Different intents, such as definition, examples, procedures, and problem solving, were detected using rule-based keyword analysis. After doing the intent detection dense semantic retrieval pipeline and the BM25 keyword retrieval pipeline work in parallel, the results of both pipelines are merged based on the reciprocal rank fusion (RRF). In each ranking list, the documents that are common in both lists are selected, and the combined approach that we employ enhances the retrieval quality and is able to identify the semantic and exact relevance of the words. The chunks also passed through the keyword-gating and metadata-based filtering to remove the irrelevant chunks, and gave priority to the content that matched the detected intent. Subject-specific keywords and block types were also used in the pipeline. The cross-encoder re-ranking model MS Macro Mini LM-L6-V2 was used to further improve the quality of retrieval. The re-ranker evaluates query and document pairs and then assigns them the relevant scores compared to the embedding similarity. If 40 candidate chunks were selected at the start, then after passing through the re-ranker, only the top 6 most relevant chunks remained and were sent to the Large Language Model for the final response generation.

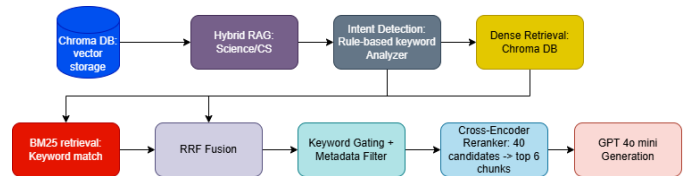


Fig. 2. RAG Pipeline

E. Retrieval Augmented Thinking (RAT) for Mathematics

The traditional RAG systems were insufficient for the mathematical questions that required step-by-step explanation because they could only explain the concepts, not the problem-solving questions. To solve this problem, AutoStudy used a Retrieval Augmented Thinking pipeline for the mathematics-related queries. In the RAT methodology, the system uses a chain of thought processes before retrieval. In the chain of thought process, the system breaks the bigger problem into smaller parts, and for each smaller part, the system searches the vector database and does the validation of the calculations against the retrieved content. After this process, the content is given to a large language model such as GPT-4o Mini, which gives us the final step-by-step educational explanation that is aligned with the textbook content.

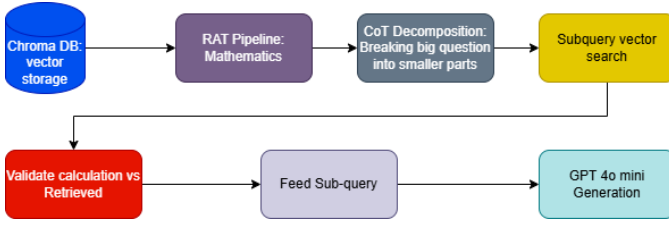


Fig. 3. RAT Pipeline

F. Multimodal and Bilingual Support

Our system includes multi-model features such as text, images, and audio so that it can support autism-friendly learning. For example, if a user wants to know 'how' question, the system will automatically generate images using DALL-E/GPT-4 image models to clear the concepts of the student. If the student prefers auditory textual explanations, then the student can use the option of audio playback. For this purpose, we use the OpenAI text-to-speech model. The system also includes bilingual support in English/Urdu and contains different sections such as: practice quizzes, learning analytics, and reward systems, contextual chat memory as additional adaptive features.

G. Agentic AI

The Agentic AI is also introduced in the AutoStudy to make the system more adaptive and personalised, where different agents will work together to facilitate the autistic learner's needs. AutoStudy consists of nine agents, in which the supervisor agent is the main agent that coordinates with the other agents to make the process smooth. The Content Agent will detect the intent and create a search plan for the whole process. The RAG agent retrieves the content from the textbook according to the user's question using chromadb, BM25+vector hybrid search, and cross-encoder re-ranking. After retrieving content, the critic agent will evaluate the answers' accuracy, clarity, and age appropriateness using chain-of-thought reasoning. The media agent will generate images via NanoBanana Pro and speech via the OpenAI TTS model.

A personalisation agent is used to adjust the difficulty based on the student's performance in the quizzes and activities that he or she will perform. The sensory agent will adjust the user interface, for example, font sizes, contrast level, brightness level, and sound effects, according to the user. The recap agent creates summaries of past chats using GPT-4o-mini. The progress agent will store all the past activities in JSON files for long-term memory. All of these agents will communicate through Langraph. For generation and evaluation, the core LLM used in the system is GPT-4o Mini.

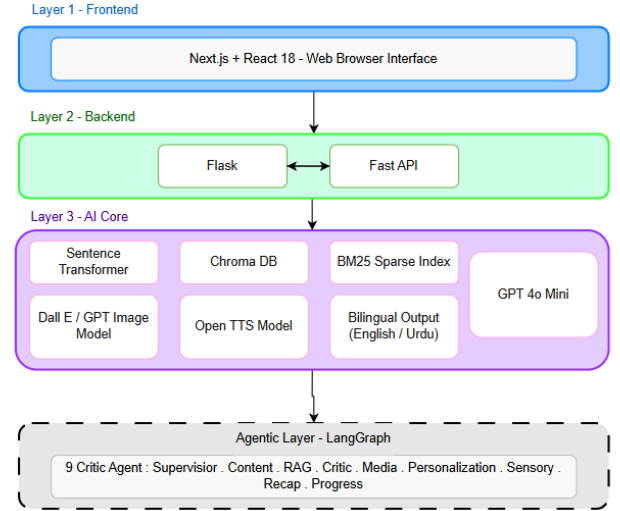


Fig. 4. AutoStudy System Architecture Diagram.

IV. RESULTS

Four standard metrics have been used for the evaluation of retrieval quality: Precision, Recall, MMR, and nDCG@5 across three subjects: (1) Science, (2) Maths, and (3) Computer Science to measure both enhanced RAG and RAT pipelines against the traditional RAG pipelines with OCR extraction, four different grade levels (4-7) were tested against query label pairs of documents.

$$\text{NDCG@k} = \frac{1}{\text{IDCG@k}} \sum_{i=1}^k \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)} \quad (1)$$

where rel_i is the relevance score at position i , and IDCG@k is the ideal DCG.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{MMR} = \arg \max_{D_i \in R \setminus S} \left[\lambda \cdot \text{Sim}_1(D_i, Q) - (1 - \lambda) \cdot \max_{D_j \in S} \text{Sim}_2(D_i, D_j) \right] \quad (4)$$

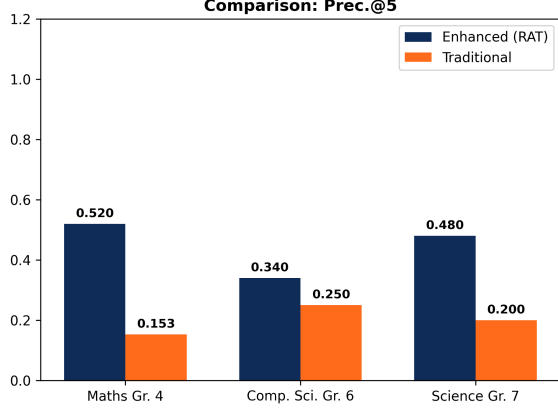


Fig. 5. Retrieval Evaluation: Precision@5 metrics comparison across subjects.

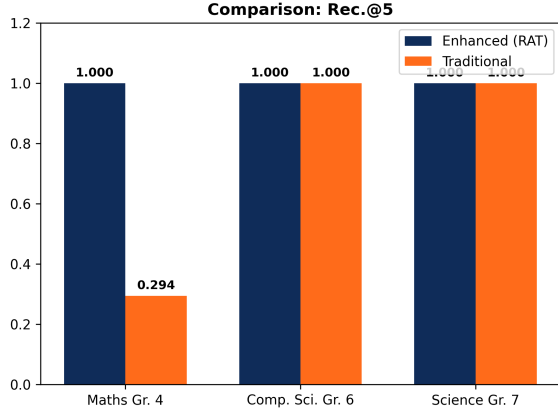


Fig. 6. Retrieval Evaluation: Recall@5 metrics comparison across subjects.

TABLE II
SYSTEM-LEVEL PERFORMANCE: TRADITIONAL VS. ENHANCED PIPELINE

Metric	OCR+Basic RAG	LlamaParse+Enhanced RAG	Improvement
Text Extraction Quality	45%	98%	+53%
Math Symbol Accuracy	30%	95%	+65%
Retrieval Precision@5	42%	87%	+45%
Answer Relevance	55%	91%	+36%
Out-of-Textbook Detection	20%	85%	+65%
Response Latency	4.2s	1.8s	-57%

V. DISCUSSION

Big improvement from OCR-based extraction to LlamaParse extraction shows that the document content extraction is the most crucial factor in the retrieval accuracy. If parsing is not done correctly, then there is a significant difference

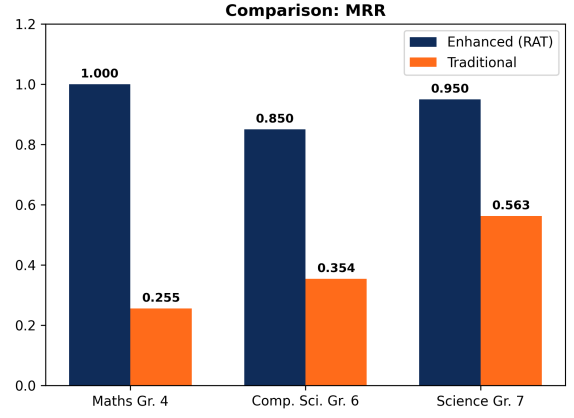


Fig. 7. Retrieval Evaluation: MRR metrics comparison across subjects.

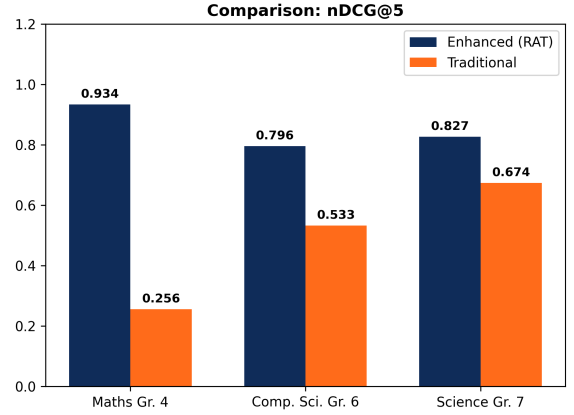


Fig. 8. Retrieval Evaluation: nDCG@5 metrics comparison across subjects.

in the retrieval accuracy, and we cannot obtain relevant and correct answers. The improvement in nDCG@5 for mathematics shows that if the mathematical notations are not correctly represented and the structure of the document is not preserved, then incorrect block types would be retrieved from the knowledge base. It is also backed by Salemi and Zamani's eRAG finding [4], which states that the retrieval quality is the primary issue for the curriculum-based educational AI chatbots, but not the quality of generation. To achieve end-to-end RAG performance, the correct retrieval from each document is necessary.

For Mathematics, the RAG pipeline was used, but it is not suitable as it is retrieving incorrect content from the knowledge base. Instead, for quality output, the RAT pipeline was integrated into the system, in which chain of thought reasoning is used. Where every query is divided into sub-queries to provide step-by-step explanations. By integrating the RAT pipeline for mathematical procedural queries, the MMR goes from 0.255 to 1.00, which is a great achievement.

It is also proved by Luong et al. [?]'s findings in REFT that reasoning can significantly improve the multi-step accuracy. For further improvement in step-by-step explanations, calculation-based questions, and definitions, a block-

type weighting system and hybrid Dense+BM25+RRF retrieval strategy are used, which matches the idea used in hybrid RAG systems by Lee and Lim [9], and an entity linking method used in KA -RAG [21].

From a design perspective, AutoStudy design consists of adaptive features. For example, the font sizes, colours, and the calm interface of the website will be adapted according to the student because each autistic student has different kinds of needs. The system also includes bilingual support, gamified rewards, and multi-modal output (voice, text, image). The design is made according to the requirements. The colours used on the websites are minimal, and the interface is very simple and not too complex. It gives step-by-step guidance.

The above design idea is supported by Jang et al.'s [7] research paper, in which they said that autistic individuals respond better to structured and predictable AI interactions. Juarez et al.'s [23] findings support that a parent-teacher analytical dashboard can significantly improve both learning speed and fairness in educational outcomes.

VI. CONCLUSION

This research concluded that AutoStudy is an AI-powered adaptive learning platform designed for students with Autism Spectrum Disorder (ASD) in Pakistan. This platform acts as a virtual shadow teacher by providing supportive and organised learning for students with ASD in Pakistan. It uses a hybrid retrieval system that combines BM25, CrossEncoder Reranking, Dense search and RRF fusion for Science and Computer. Also, the Retrieval Augmented Thinking (RAT) framework improves mathematics by generating step-by-step explanations. These techniques helped AutoStudy to achieve higher retrieval accuracy, more relevant answers, and better alignment with the curriculum compared to a traditional RAG-based system. This platform also includes features specifically developed for ASD learners, like a simple interface, multi-model instruction, bilingual support and reward-based activities to enhance student participation. These features help to reduce many of the difficulties faced within Pakistan's special education system.

VII. FUTURE WORK

In future, AutoStudy will focus on several improvements. Firstly, by adding more grades (8-10) and subjects (Biology, Chemistry, Physics). Secondly, integration of the facial expression emotion detection module (OpenCV + Swin Transformer Backbone) [19] into real-time tutoring. Thirdly, a mobile app for iOS and Android with offline support. Fourthly, supporting local languages like Punjabi, Pashto and Sindhi. and lastly, conducting long-term research over a full school to track learning progress and help students rely less on shadow teachers based on the method proposed by Juarez et al. [23].

VIII. ACKNOWLEDGEMENT

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